Can machine learning techniques find an edge in Irish greyhound racing win markets?

Research Question

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Abstract

This study examines whether machine learning techniques can be used to identify a profitable betting strategy in Irish greyhound racing win markets. Historical race data was collected from the Greyhound Racing Ireland website, covering 74,927 races between 18 June 2020 and 31 March 2026. The final dataset included race-level, runner-level, and historical performance information, from which 62 predictive features were constructed. A LightGBM model was trained using a chronological split of the data into training, validation, and test sets in order to reflect a realistic prediction setting. The model's output probabilities were calibrated using isotonic regression and then normalised within each race to produce final win probabilities. These probabilities were used to identify positive expected value betting opportunities against bookmaker starting prices. The strongest outcome was achieved using the model's top picks with isotonic-calibrated and normalised probabilities, where the identified strategy produced 504.95% bankroll growth under a constrained Kelly staking approach. Overall, the findings suggest that machine learning can provide an edge in Irish greyhound racing when combined with probability calibration, selective filtering, and disciplined staking.

1. Introduction

There are approximately 300 greyhound races a week across 17 tracks in Ireland. The prize money for these races ranges from just a couple hundred euro to potentially six figure sums. The Irish derby ran in Shelbourne Park every year in early autumn boasts a €300,000 prize fund, with the winning owner taking home a staggering €125,000. However, there is a much more lucrative side to greyhound racing, the betting markets.

This research uses machine learning and data analytics techniques to find a profitable betting strategy for greyhound races in Ireland. The data used in this research was collected from the Greyhound Racing Ireland website and contains over 93,000 distinct races between 18th of June 2020 to 31st March 2026. The outcome of the model isn't to just predict what dog will finish

first in the race, but to use the probability outputted to determine the fair odds that should be offered and compare that to the starting price (SP) offered by bookmakers on the day of the race to determine if a positive expected value opportunity exists.

A lot of the previous work done in this area only looks at the model's top pick accuracy. Given the high levels of uncertainty in greyhound races, I will not rely solely on the top pick accuracy, but rather the calibration of the models outputted probabilities in order to find an edge.

2. Related Works

2.1 Literature on Sports Betting Markets and Value Based Wagering

There is a large amount of research out there for sports predictions and betting strategies. Kelly (1956) originally proposed a growth-optimal staking rule based on maximizing the expected logarithm of wealth. The criterion was later adapted to practical gambling applications by Thorp (1969), who applied and popularised it in blackjack and other favourable betting settings. Value-based wagering is based on identifying bets where the bettor's estimated probability of an outcome is greater than the probability implied by the bookmaker's odds, creating positive expected value. This highlights that in sports betting, the quality and calibration of probability estimates matters more than just the models top pick accuracy (Walsh and Joshi, 2024).

2.2 Literature on Machine learning techniques in Greyhound racing or similar

Chen et al (1994) used a form of decision tree (ID3) to predict race winners. They then used a flat staking technique of $2 per bet on the models top pick, as well as the pick from an expert. Their model correctly predicted 34 winners, 50 losers and no selection in 26 races out of 100. They're experts chose 19 winners and 81 losers. Subsequently they're flat staking strategy led to a profit when using the models picks and a loss with the experts picks.

Johansson and Sönströd (2003) used a neural network and a similar flat stake betting strategy to Chen. They kept their feature dataset simple and the output was binary classification of winner 1 or loser 0. Their strategy actually resulted in a loss of 3% when only looking at win markets despite having a 24.9% accuracy rate. This shows the complexity of determining a profitable strategy.

Schumaker and Johnson (2008) explored the use of Support Vector Regression (SVR) in greyhound racing, using historical race variables to predict finishing performance and test betting strategies. They achieved a 45.35% win rate with their strategy, which involved betting on a number of dogs in the same race.

Lyons (2016) showed that richer feature engineering may lead to improved prediction, but machine learning performance still remained below an expert benchmark. This suggests that success depends not only on feature design, but also on model choice and how race outcomes are framed.

Woo, So and Lee (2025)[1] examined gradient-boosted decision tree models in a horse-race prediction context and reported that, while CatBoost achieved the highest level of predictive accuracy, LightGBM provided a more efficient alternative and performed better in betting scenarios. This is relevant to the present study because greyhound racing data is also highly tabular and shaped by complex interactions between variables. As a result, LightGBM appears to be an appropriate modelling approach where both predictive performance and practical usability are important.

3. Methodology

3.1 Data Mining

All data used in this study was collected from Greyhound Racing Ireland's official website. A Python script was developed to scrape historical race cards from the site. Each race card typically contains 10 to 12 races. As much relevant information as possible was captured from each race. In addition to race-level and runner-level data, the script also collected the full historical form of every greyhound that appeared in the sampled races.

When constructing features, only information that would have been available before the date of the race was used. This was important in order to avoid look-ahead bias and to ensure that the model reflected a realistic pre-race prediction setting.

[1] This paper was translated from Korean with the help of Gen AI

For consistency, the study focused only on middle-distance races. This is indicated through the grade code, where grades prefixed with A (example A7) represent middle distance races, while codes such as S0–S10 represent sprint races. Only races with an A-grade classification were included in the dataset. This was applied to improve comparability across observations and to reduce variation in variables that arose from different race types.

In total, the dataset included 74,927 races between 18th June 2020 and 31st March 2026, involving 30,588 unique greyhounds. 815,749 rows of historical form data was gathered. As a result, the final dataset contained information at three levels: race-level variables, runner-level variables, and performance-level variables derived from each dog's historical form.

3.2 Pre-processing

Several pre-processing decisions were made before model development. First, starting price (SP) was excluded from the training dataset. The main reason for this was that SP is not known with certainty before the race begins and therefore would reduce the realism of the prediction task. In addition, SP reflects market sentiment, and including it could cause the model to rely too heavily on bookmaker expectations rather than identifying patterns directly from the underlying racing data. Excluding SP therefore made the model a purer test of whether race outcomes could be predicted from race and form variables alone.

Trial runs were excluded when feature engineering because they do not reflect the same competitive conditions as official races, and therefore led to instances where there was missing race performance metrics.

Pre-processing was also guided by the competitive nature of greyhound racing. Bolton and Chapman (1986) argue that race prediction should be treated as a within-race competitive process rather than as a set of isolated individual observations. This provides justification for the inclusion of race-relative features, which measure a dog's characteristics in relation to the other runners in the same race. For example, rather than only including a greyhound's average sectional time as a variable, the dataset also included the greyhound's ranked average sectional time within that race field. This was intended to capture the relative competitive position of each runner more effectively. Similar race-relative transformations were applied to other variables where appropriate, as this better reflects the fact that winning depends not only on a dog's own ability, but on how that ability compares with the opposition in that specific race.

3.3 Feature Selection and Model Choice

The final model used 62 features, many of which were derived from historical performance data and race-level details. These included variables related to past form, trap information, race characteristics, venue indicators, and comparative features within each race. A relatively broad feature set was used to reflect the fact that greyhound race outcomes are influenced by multiple interacting factors.

LightGBM was selected because it is an efficient gradient boosting framework that performs well on large tabular datasets (Ke et al., 2017). This made it suitable for the present study, which used a large volume of structured runner-level, race-level, and historical data. LightGBM is also well suited to capturing non-linear relationships and interactions between variables, which is important in greyhound racing where outcomes depend on factors such as trap position, grade, sectional speed, and recent form.

3.4 Training and Testing

The model was trained and evaluated using a chronological race-level split in order to reflect a realistic prediction setting and prevent leakage between runners in the same race. Races were ordered by meeting date and divided into three contiguous groups: 60% training, 20% validation, and 20% testing. LightGBM was first trained on the training set only. The validation set was then used for model tuning and probability calibration, while final model performance was assessed on the unseen test set.

3.5 Probability Calibration

Following LightGBM prediction, the raw output probabilities were treated as independent predicted win probabilities for each runner. These were calibrated using isotonic regression on the validation set in order to improve alignment between predicted probabilities and observed outcomes. The calibrated probabilities within each race did not sum to one as they should have since the model predicts runners independently. Therefore they were normalised within each race so that the probabilities across all runners summed to one. These probabilities were then used for ranking and evaluation.

3.6 Evaluation metrics

Model performance was evaluated using a combination of statistical prediction metrics and betting performance metrics. Standard measures like ROC-AUC, Brier score, precision, recall, F1-score, and top-pick accuracy, were used to assess how well the model distinguished winners from non-winners and how well the predicted probabilities aligned with actual outcomes.

However, the main aim of this research was not simply to maximise predictive accuracy, but to identify whether the model could be used to find a profitable betting strategy. For that reason, the most important evaluation measures were return on investment (ROI) and total bankroll growth when the model's estimated probabilities were compared with market odds to identify value bets. These metrics were used to assess the practical usefulness of the model in a betting context.

4. Results

4.1 Overview of Test Performance

The model was evaluated on the most recent races in the held-out test set. LightGBM achieved a ROC-AUC of 0.6562 and a top-pick accuracy of 32.91%. Calibration improved after isotonic regression, with the Brier score falling from 0.2229 to 0.1349, suggesting the adjusted probabilities were more reliable.

4.2 Classification Performance

The classification report showed an overall accuracy of 62.7% on the test set. However, this figure should be interpreted cautiously, as race prediction is a highly imbalanced problem in which most runners do not win. For the winner class, precision was 0.247, recall was 0.586, and the F1-score was 0.348. This indicates that the model was able to identify a reasonable share of actual winners, although with a substantial number of false positives. Therefore, simple classification accuracy alone is not an adequate measure of practical betting usefulness.

4.3 Ranking Performance

Since the model was ultimately used to rank runners within each race, top-pick accuracy was a more meaningful metric than overall classification accuracy. The highest-rated runner in each race won in 32.91% of test races (4,932 wins from 14,985 races). This indicates that the model was able to identify the winner more often than chance (~16.67% for a 6 dog race) and more often than a greyhound expert at Shelbourne Park with a 23.7% accuracy, although over a different time period (Lyons, 2016).

4.4 Probability Calibration

Probability calibration improved materially after post-processing. The raw LightGBM probabilities produced a Brier score of 0.2229, whereas the isotonic-calibrated probabilities achieved a Brier score of 0.1349. This suggests that isotonic regression improved the alignment between predicted win probabilities and actual outcomes. This is particularly important in a betting context, as profitable value betting depends not only on identifying likely winners, but on producing probability estimates that are as realistic as possible relative to market prices.

4.5 Betting Performance

To assess the practical usefulness of the model, a betting strategy was evaluated using Kelly Criterion staking. Bets were placed when the model's estimated probability implied a positive expected value relative to the market SP. Two strategies were examined. The first placed bets on all selections that met the positive expected value condition. The second imposed an additional restriction, requiring that the selection was also the model's top-ranked runner in that race.

Minimum Edge	Number of Bets	ROI
0%	13,027	-10.35%
5%	10,649	-10.32%
10%	8,766	-9.88%
20%	5,913	-8.22%
30%	4,013	-5.99%
50%	2,014	-2.60%
60%	1,437	1.55%
70%	1,047	-0.47%

Table 1: All positive expected value selections

Minimum Edge	Number of Bets	ROI
0%	5,044	4.15%
5%	4,212	4.04%
10%	3,522	4.51%
20%	2,424	6.82%
30%	1,638	9.88%
50%	791	16.70%
60%	536	21.30%
70%	386	21.80%

Table 2: Top-pick only positive expected value selections

When all positive expected value selections were backed, the strategy generated a negative ROI at almost every minimum edge threshold, with the exception of the 60% edge level. This suggests that although the model identified a large number of apparent value opportunities, many of these were not sufficiently strong in practice to overcome estimation error and the volatility introduced by proportional Kelly staking.

Restricting bets to selections that were both positive expected value opportunities and the model's top pick produced consistently positive ROI figures across all minimum edge thresholds shown. This indicates that the model's strongest signals were more reliable than weaker value signals further down the ranking. In practical terms, the top-pick filter improved selection quality by concentrating only on the model's most strongly supported predictions.

As the minimum edge threshold increases, the number of bets falls, while ROI generally rises. This is intuitive, as higher edge requirements produce fewer but more selective betting opportunities. However, higher thresholds also reduce betting frequency, which may limit the extent to which compounding can operate over time.

To explore the long-run performance of the strategy, simulations were conducted at a 60% minimum edge threshold. This threshold was selected because it provided both a relatively high ROI and a sufficiently large number of bets to allow compounding effects to emerge. The simulations tested different fractions of Kelly stakes and maximum stake caps as a proportion of bankroll. These controls were included to avoid the bankroll being depleted before the strategy's edge could materialise over a longer sequence of bets.

Kelly Fraction/Kelly Cap	0.01	0.025	0.05	0.1	0.2	0.3
0.125	119.03%	257.78%	381.88%	381.88%	381.88%	381.88%
0.25	119.03%	297.63%	299.46%	504.95%	504.95%	504.95%
0.5	119.03%	294.62%	177.81%	-69.99%	-58.50%	-58.50%
0.75	119.03%	294.62%	173.52%	-97.47%	-99.93%	-99.90%
1	119.03%	294.62%	173.52%	-97.72%	-100.00%	-100.00%

Table 3: Simulated bankroll growth by Kelly fraction and maximum stake cap top-pick only

The simulation results show that bankroll growth was highly sensitive to staking aggressiveness. Where no Kelly reduction was applied and the maximum stake cap was set too high, the strategy produced extremely large drawdowns and, in some scenarios, an almost complete loss of bankroll before any long-run edge could be realised. In contrast, more conservative staking rules generated substantially stronger outcomes. The best-performing scenario was a quarter Kelly strategy with a 10% bankroll cap per bet, which produced 504.95% bankroll growth over the sample period.

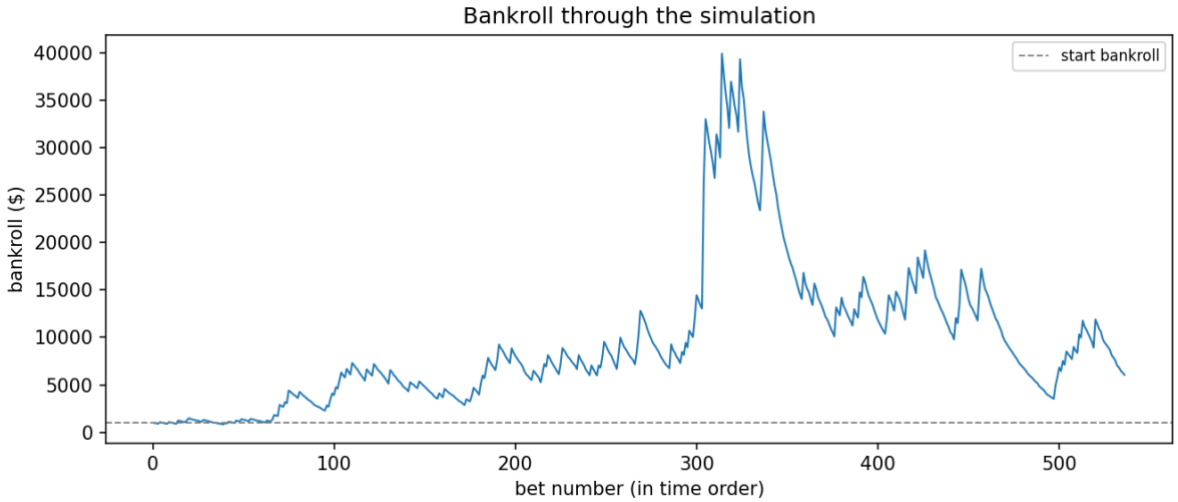

Figure 1: Simulated bankroll growth over time top-pick only

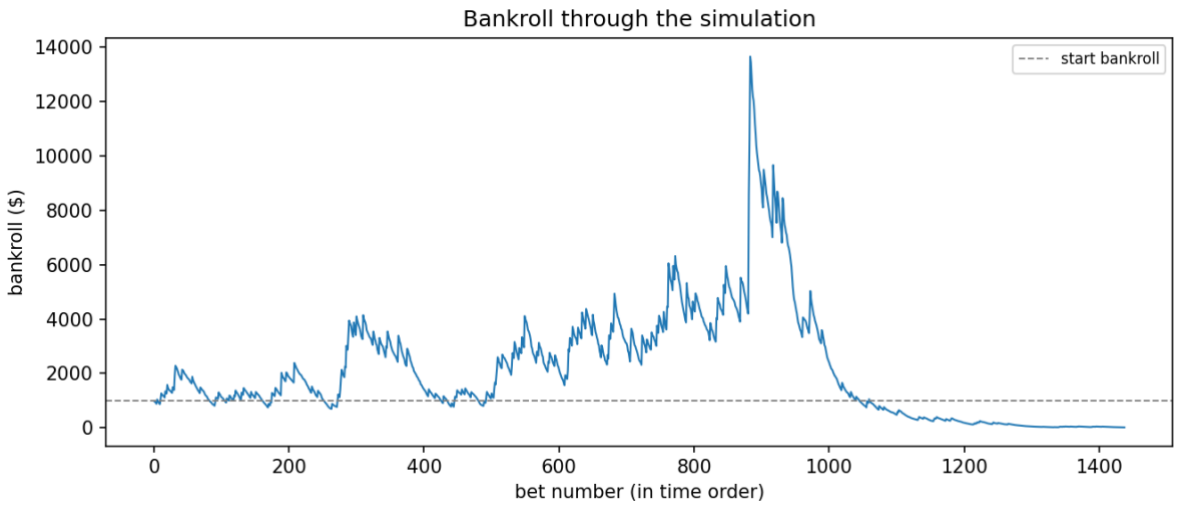

Figure 2: Simulated bankroll growth over time all selections

The bankroll path highlights the volatility of the strategy. Starting from an initial bankroll of $1,000, the simulated bankroll rose to approximately $40,000 before declining sharply but ending above $6,000. A similar pattern was observed when all valid selections were included, with the bankroll rising to just under $14,000 before eventually declining to nearly $0. This shows the potential profitability of the approach and the considerable volatility associated with greyhound betting, particularly when returns are magnified through compounding. Overall, the findings suggest that the model may have practical betting value, but only when paired with selective bet filtering and disciplined staking constraints.

5. Conclusion

This study set out to examine whether machine learning techniques could be used to identify a profitable betting strategy in Irish greyhound racing win markets. The results suggest that they can, but only under specific conditions. While the LightGBM model showed moderate predictive performance overall, its main value came from producing probability estimates that could be compared with bookmaker prices to identify potential value opportunities. The findings showed that backing all apparent positive expected value selections was generally not profitable. However, when betting was restricted to the model's top-ranked runners, the strategy produced positive ROI across the tested edge thresholds, with the strongest simulated outcome generating 504.95% bankroll growth under a constrained Kelly staking approach.

A further limitation of the study is that some potentially useful information only becomes available very shortly before the race. For example, a greyhound's weight is recorded just prior to the start and may provide an indication of the dog's condition on the day. This type of late information was not included in the model, but may already have been incorporated into the starting price by the betting market. As a result, bookmaker odds may have drifted and been more reflective of the true probability than the model's estimates, particularly for runners ranked further down the model's order. This may help explain why broader positive expected value strategies performed poorly compared with the more selective top-pick approach.

This research provides evidence that machine learning can be used to identify a profitable greyhound betting strategy, but that success depends on how the model outputs are translated into practical betting decisions. Further research could explore additional late-stage variables, hyper-parameter tuning, stricter feature selection, alternative models such as Neural Networks or SVM, and other betting markets such as forecasts or top finishes to test whether performance can be improved further.

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Gen-AI Declaration

Gen-AI was used in my research. I used Gen-AI to assist with:

- Identifying academic papers and research relevant to my research topic
- Improving and optimising code. Given the large dataset I was operating with the code needed to be optimised for efficiency. Even the optimised code required me to run the

script overnight in order to build the full feature dataset. The run-time on my machine was over 1hour and 40 minutes for this task highlighting the necessity for optimization here.

- Summarising key points in order to adhere to the word count.